Lecture Notes on Finite-Field Matrices, Full Rank, Parity-Check Maps, and Syndromes

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1 Overview

These notes summarize a basic but important story in linear algebra over finite fields and its application to coding theory.

The main points are:

- Counting all matrices over a finite field.
- Counting invertible matrices over a finite field.
- Computing the probability that a random matrix is invertible or full rank.
- Understanding parity-check matrices as linear maps.
- Interpreting the syndrome map as a surjective map whose kernel is the code.
- Packaging the whole picture as a short exact sequence

$$0 \longrightarrow \mathcal{C} \longrightarrow \mathbb{F}_q^n \xrightarrow{H} \mathbb{F}_q^{n-k} \longrightarrow 0.$$

Throughout, q is a prime power and $\mathbb{F}_q$ denotes the finite field with q elements.

2 Counting Matrices over $\mathbb{F}_q$

2.1 All matrices

Proposition 2.1. *The number of $m \times n$ matrices over $\mathbb{F}_q$ is*

$$\left|\mathbb{F}_q^{m\times n}\right| = q^{mn}.$$

In particular,

$$\left|\mathbb{F}_q^{m\times m}\right| = q^{m^2}.$$

Proof. An $m \times n$ matrix has mn entries. Each entry can be chosen independently from $\mathbb{F}_q$, which has q elements. Therefore the total number of matrices is q^{mn}. □

Remark 2.2. This is the simplest counting principle in the subject: finite-field matrix spaces are just finite Cartesian products.

3 Counting Invertible Matrices

3.1 The general linear group

Definition 3.1. The *general linear group* over $\mathbb{F}_q$ is

$$\mathrm{GL}_m(\mathbb{F}_q) = \{A \in \mathbb{F}_q^{m\times m} : A \text{ is invertible}\}.$$

Theorem 3.2. *The number of invertible $m \times m$ matrices over $\mathbb{F}_q$ is*

$$|\,\mathrm{GL}_m(\mathbb{F}_q)| = (q^m - 1)(q^m - q)(q^m - q^2)\cdots(q^m - q^{m-1}).$$

Equivalently,

$$|\,\mathrm{GL}_m(\mathbb{F}_q)| = \prod_{i=0}^{m-1}(q^m - q^i).$$

Proof. We count invertible matrices by choosing their columns.

A matrix is invertible if and only if its columns form a basis of $\mathbb{F}_q^m$, equivalently if and only if its columns are linearly independent.

- The first column can be any nonzero vector of $\mathbb{F}_q^m$, so there are $q^m - 1$ choices.
- The second column must not lie in the span of the first column. Since the span of one nonzero vector has q elements, there are $q^m - q$ choices.
- The third column must not lie in the span of the first two columns. If those are independent, their span has q^2 elements, so there are $q^m - q^2$ choices.
- Continuing this way, the k-th column must avoid a subspace of size q^{k-1}, so there are $q^m - q^{k-1}$ choices.

Multiplying all factors gives

$$|\,\mathrm{GL}_m(\mathbb{F}_q)| = \prod_{i=0}^{m-1}(q^m - q^i).$$

□

Remark 3.3. An invertible matrix is the same thing as an *ordered basis* of $\mathbb{F}_q^m$, written as columns. Thus

$$|\,\mathrm{GL}_m(\mathbb{F}_q)| = \#\{\text{ordered bases of } \mathbb{F}_q^m\}.$$

3.2 A useful factorization

Proposition 3.4. *One may rewrite the above formula as*

$$|\,\mathrm{GL}_m(\mathbb{F}_q)| = q^{m^2}\prod_{j=1}^{m}(1 - q^{-j}).$$

Equivalently,

$$|\,\mathrm{GL}_m(\mathbb{F}_q)| = q^{\frac{m(m-1)}{2}}\prod_{j=1}^{m}(q^j - 1).$$

Proof. Starting from

$$|\operatorname{GL}_m(\mathbb{F}_q)| = \prod_{i=0}^{m-1} (q^m - q^i),$$

factor out q^m from each term:

$$q^m - q^i = q^m(1 - q^{i-m}).$$

Hence

$$|\operatorname{GL}_m(\mathbb{F}_q)| = q^{m^2} \prod_{i=0}^{m-1} (1 - q^{i-m}).$$

Reindexing with $j = m - i$ yields

$$|\operatorname{GL}_m(\mathbb{F}_q)| = q^{m^2} \prod_{j=1}^{m} (1 - q^{-j}).$$

For the second factorization, write

$$q^m - q^i = q^i(q^{m-i} - 1),$$

so

$$\prod_{i=0}^{m-1} (q^m - q^i) = \left(\prod_{i=0}^{m-1} q^i\right) \left(\prod_{i=0}^{m-1} (q^{m-i} - 1)\right) = q^{0+1+\cdots+(m-1)} \prod_{j=1}^{m} (q^j - 1),$$

and

$$0 + 1 + \cdots + (m-1) = \frac{m(m-1)}{2}.$$

□

4 Probability that a Random Square Matrix is Invertible

4.1 Exact formula

A uniformly random $m \times m$ matrix over $\mathbb{F}_q$ is invertible with probability

$$\mathbb{P}(A \in \operatorname{GL}_m(\mathbb{F}_q)) = \frac{|\operatorname{GL}_m(\mathbb{F}_q)|}{|\mathbb{F}_q^{m\times m}|}.$$

Since

$$|\mathbb{F}_q^{m\times m}| = q^{m^2},$$

we obtain the following.

Theorem 4.1. *For a uniformly random matrix $A \in \mathbb{F}_q^{m\times m}$,*

$$\mathbb{P}(A \textit{ invertible}) = \prod_{j=1}^{m} (1 - q^{-j}).$$

Therefore

$$\mathbb{P}(A \textit{ singular}) = 1 - \prod_{j=1}^{m} (1 - q^{-j}).$$

Proof. Divide the formula for $|\operatorname{GL}_m(\mathbb{F}_q)|$ by q^{m^2}. □

4.2 Row-by-row viewpoint

There is another derivation that matches Gaussian elimination more closely.

Proposition 4.2. *Let $A \in \mathbb{F}_q^{m\times m}$ be chosen uniformly at random. Then*

$$\mathbb{P}(A \textit{ invertible}) = \prod_{k=1}^{m} \left(1 - q^{k-1-m}\right) = \prod_{j=1}^{m} (1 - q^{-j}).$$

Proof. Write the rows of A as $r_1, \dots, r_m \in \mathbb{F}_q^m$. The matrix is invertible if and only if its rows are linearly independent.

- r_1 must be nonzero, which occurs with probability
$$\frac{q^m - 1}{q^m} = 1 - q^{-m}.$$
- Given that $r_1 \neq 0$, the row r_2 must avoid $\operatorname{span}(r_1)$, which has q elements. Thus the conditional probability is
$$\frac{q^m - q}{q^m} = 1 - q^{1-m}.$$
- Given that r_1, r_2 are independent, the row r_3 must avoid their span, which has q^2 elements. Thus
$$\frac{q^m - q^2}{q^m} = 1 - q^{2-m}.$$

Continuing in this manner, the k-th row must avoid a subspace of size q^{k-1}, so the corresponding conditional probability is

$$1 - q^{k-1-m}.$$

Multiplying all conditional probabilities yields

$$\mathbb{P}(A \text{ invertible}) = \prod_{k=1}^{m} (1 - q^{k-1-m}).$$

Reindexing gives

$$\prod_{j=1}^{m} (1 - q^{-j}).$$

□

Remark 4.3. This is exactly the condition that Gaussian elimination finds a pivot in every row.

4.3 Examples

Example 4.4. For $m = 2$,

$$|\operatorname{GL}_2(\mathbb{F}_q)| = (q^2 - 1)(q^2 - q),$$

so

$$\mathbb{P}(A \in \operatorname{GL}_2(\mathbb{F}_q)) = (1 - q^{-1})(1 - q^{-2}).$$

If $q = 2$, then

$$\mathbb{P}(A \in \operatorname{GL}_2(\mathbb{F}_2)) = \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{4}\right) = \frac{3}{8}.$$

Indeed,

$$|\operatorname{GL}_2(\mathbb{F}_2)| = (4 - 1)(4 - 2) = 6,$$

and $6/16 = 3/8$.

Example 4.5. For $m = 3$ and $q = 2$,

$$\mathbb{P}(A \in \mathrm{GL}_3(\mathbb{F}_2)) = \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{8}\right) = \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{7}{8} = \frac{21}{64}.$$

Since $|\mathbb{F}_2^{3\times 3}| = 2^9 = 512$, it follows that

$$|\,\mathrm{GL}_3(\mathbb{F}_2)| = 512 \cdot \frac{21}{64} = 168.$$

5 Rectangular Matrices and Full Row Rank

In coding theory, parity-check matrices are generally rectangular rather than square.

5.1 Counting full-row-rank matrices

Let $H \in \mathbb{F}_q^{r\times n}$ with $r \leq n$. We want to count those matrices having full row rank r.

Theorem 5.1. *The number of $r \times n$ matrices over $\mathbb{F}_q$ having full row rank r is*

$$\prod_{i=0}^{r-1}(q^n - q^i).$$

Proof. Choose the rows one by one so that they remain linearly independent.

- The first row can be any nonzero vector in $\mathbb{F}_q^n$, so there are $q^n - 1$ choices.
- The second row must avoid the span of the first, which has q elements, so there are $q^n - q$ choices.
- The third row must avoid the span of the first two, which has q^2 elements, so there are $q^n - q^2$ choices.

Continuing, the r-th row must avoid a subspace of size q^{r-1}, giving $q^n - q^{r-1}$ choices. Multiplying yields

$$\prod_{i=0}^{r-1}(q^n - q^i).$$

□

5.2 Probability of full row rank

Since the total number of $r \times n$ matrices is q^{rn}, we obtain:

Corollary 5.2. *A uniformly random matrix $H \in \mathbb{F}_q^{r\times n}$ has full row rank r with probability*

$$\mathbb{P}(\mathrm{rank}(H) = r) = \frac{\prod_{i=0}^{r-1}(q^n - q^i)}{q^{rn}} = \prod_{i=0}^{r-1}(1 - q^{i-n}).$$

Example 5.3. Let $H \in \mathbb{F}_2^{3\times 7}$ be uniformly random. Then

$$\mathbb{P}(\mathrm{rank}(H) = 3) = \left(1 - 2^{-7}\right)\left(1 - 2^{-6}\right)\left(1 - 2^{-5}\right).$$

That is,

$$\mathbb{P}(\mathrm{rank}(H) = 3) = \frac{127}{128} \cdot \frac{63}{64} \cdot \frac{31}{32} = \frac{248031}{262144} \approx 0.946.$$

So a random 3×7 binary matrix is full rank with probability about 94.6%.

6 Linear Codes and Parity-Check Matrices

6.1 Linear codes

Definition 6.1. An $[n, k]$ *linear code over* $\mathbb{F}_q$ is a k-dimensional linear subspace

$$\mathcal{C} \subseteq \mathbb{F}_q^n.$$

Remark 6.2. The parameter n is the block length, and k is the dimension of the code. Thus $|\mathcal{C}| = q^k$.

6.2 Parity-check matrices

Definition 6.3. A matrix $H \in \mathbb{F}_q^{(n-k)\times n}$ is called a *parity-check matrix* for a linear code $\mathcal{C} \subseteq \mathbb{F}_q^n$ if

$$\mathcal{C} = \mathrm{Ker}(H),$$

where H is viewed as the linear map

$$H : \mathbb{F}_q^n \to \mathbb{F}_q^{n-k}, \qquad x \longmapsto Hx^\top.$$

Remark 6.4. The parity-check matrix describes the code as the solution space of a system of linear equations. A word $x \in \mathbb{F}_q^n$ belongs to the code precisely when it satisfies all parity checks:

$$Hx^\top = 0.$$

7 Full Rank and Surjectivity of the Syndrome Map

7.1 The syndrome map

Given a parity-check matrix $H \in \mathbb{F}_q^{(n-k)\times n}$, define the associated linear map

$$H : \mathbb{F}_q^n \to \mathbb{F}_q^{n-k}, \qquad x \mapsto Hx^\top.$$

This is often called the *syndrome map*.

Definition 7.1. For any $y \in \mathbb{F}_q^n$, the *syndrome* of y is

$$s(y) = Hy^\top \in \mathbb{F}_q^{n-k}.$$

7.2 Full row rank is equivalent to surjectivity

Theorem 7.2. *Let $H \in \mathbb{F}_q^{(n-k)\times n}$. Then the following are equivalent:*

(1) *H has full row rank:*

$$\mathrm{rank}(H) = n - k.$$

(2) *The linear map*

$$H : \mathbb{F}_q^n \to \mathbb{F}_q^{n-k}$$

is surjective.

Proof. Since the image of a linear map has dimension equal to the rank of its matrix,

$$\dim(\mathrm{Im}\, H) = \mathrm{rank}(H).$$

Thus $\mathrm{rank}(H) = n - k$ if and only if

$$\dim(\mathrm{Im}\, H) = n - k.$$

But $\mathrm{Im}\, H$ is a subspace of $\mathbb{F}_q^{n-k}$, whose dimension is also $n - k$. Therefore $\dim(\mathrm{Im}\, H) = n - k$ if and only if

$$\mathrm{Im}\, H = \mathbb{F}_q^{n-k},$$

which is exactly surjectivity. □

7.3 Rank-nullity and the code dimension

Corollary 7.3. *If $H \in \mathbb{F}_q^{(n-k)\times n}$ has full row rank $n-k$, then*

$$\dim \operatorname{Ker}(H) = k.$$

Proof. By rank-nullity,

$$\dim \operatorname{Ker}(H) + \operatorname{rank}(H) = n.$$

If $\operatorname{rank}(H) = n-k$, then

$$\dim \operatorname{Ker}(H) = n - (n-k) = k.$$

□

Remark 7.4. This is exactly why one wants a parity-check matrix to have full row rank: otherwise some parity checks are redundant, and the kernel has dimension larger than intended.

8 The Meaning of the Syndrome

8.1 Syndrome as measurement of parity-check failure

For a vector $y \in \mathbb{F}_q^n$, the syndrome

$$Hy^\top$$

records the values of all parity-check equations evaluated on y.

- If $Hy^\top = 0$, then $y \in \mathcal{C}$.
- If $Hy^\top \neq 0$, then $y \notin \mathcal{C}$, and the syndrome measures how the parity checks fail.

Thus the syndrome is not a codeword; rather, it is the image of a word under the parity-check map.

8.2 Syndrome of a received word with error

Suppose $c \in \mathcal{C}$ is a codeword and $e \in \mathbb{F}_q^n$ is an error vector. If the received word is

$$y = c + e,$$

then

$$Hy^\top = H(c+e)^\top = Hc^\top + He^\top = 0 + He^\top = He^\top.$$

So:

Proposition 8.1. *If $y = c + e$ with $c \in \mathcal{C}$, then the syndrome of y depends only on the error e:*

$$s(y) = Hy^\top = He^\top.$$

Remark 8.2. This is the foundation of syndrome decoding. The syndrome does not depend on which codeword was sent; it depends only on the error pattern modulo the code.

9 Syndrome as a Coset Label

9.1 Cosets of the code

Since $\mathcal{C} = \mathrm{Ker}(H)$ is a subspace of $\mathbb{F}_q^n$, we may consider cosets

$$v + \mathcal{C} = \{v + c : c \in \mathcal{C}\}.$$

Proposition 9.1. *Two vectors $x, y \in \mathbb{F}_q^n$ have the same syndrome if and only if they lie in the same coset of $\mathcal{C}$.*

Proof. We have

$$Hx^\top = Hy^\top \iff H(x-y)^\top = 0 \iff x - y \in \mathrm{Ker}(H) = \mathcal{C}.$$

This is equivalent to saying that x and y lie in the same coset modulo $\mathcal{C}$. □

Corollary 9.2. *The syndrome is a complete label for the cosets of $\mathcal{C}$ in $\mathbb{F}_q^n$.*

Proof. The previous proposition shows that the fibers of the syndrome map are exactly the cosets of $\mathcal{C}$. If H is surjective, every syndrome occurs, so the set of syndromes is in bijection with the set of cosets. □

9.2 Number of cosets

If H has full row rank $n - k$, then

$$\dim(\mathcal{C}) = k.$$

Hence

$$|\mathcal{C}| = q^k, \qquad |\mathbb{F}_q^n| = q^n.$$

Therefore the number of cosets is

$$\frac{q^n}{q^k} = q^{n-k}.$$

But this is also the number of syndrome vectors in $\mathbb{F}_q^{n-k}$.

So the syndrome map gives a natural identification:

$$\mathbb{F}_q^n / \mathcal{C} \cong \mathbb{F}_q^{n-k}.$$

10 Short Exact Sequence Viewpoint

10.1 The fundamental exact sequence

When $H \in \mathbb{F}_q^{(n-k)\times n}$ has full row rank, the code $\mathcal{C} = \mathrm{Ker}(H)$ fits into the short exact sequence

$$0 \longrightarrow \mathcal{C} \longrightarrow \mathbb{F}_q^n \xrightarrow{H} \mathbb{F}_q^{n-k} \longrightarrow 0.$$

Let us unpack this.

- The map $\mathcal{C} \to \mathbb{F}_q^n$ is the inclusion, hence injective.
- The kernel of $H : \mathbb{F}_q^n \to \mathbb{F}_q^{n-k}$ is exactly $\mathcal{C}$.
- Since H has full row rank, it is surjective.

This is exactly what exactness means.

10.2 Interpretation

The short exact sequence says:

- $\mathcal{C}$ consists of words that satisfy all parity checks.
- $\mathbb{F}_q^n$ is the ambient space of all words.
- $\mathbb{F}_q^{n-k}$ is the space of all possible syndrome values.

So the quotient of the ambient space by the code is measured by syndromes:

$$\mathbb{F}_q^n/\mathcal{C} \cong \mathbb{F}_q^{n-k}.$$

Remark 10.1. This is the clean algebraic meaning of the slogan:

$$\text{syndrome} = \text{coset label}.$$

11 A Concrete Binary Example

11.1 A 3×7 parity-check matrix

Consider

$$H = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix} \in \mathbb{F}_2^{3\times 7}.$$

Its rows are linearly independent, so

$$\operatorname{rank}(H) = 3.$$

Therefore the map

$$H : \mathbb{F}_2^7 \to \mathbb{F}_2^3$$

is surjective.

Let

$$\mathcal{C} = \operatorname{Ker}(H).$$

Then rank-nullity gives

$$\dim(\mathcal{C}) = 7 - 3 = 4.$$

Thus $\mathcal{C}$ is a $[7, 4]$ binary linear code.

11.2 Syndromes

Since the codomain is $\mathbb{F}_2^3$, there are exactly $2^3 = 8$ possible syndromes. Because H is surjective, all 8 syndrome vectors occur.

The cosets of $\mathcal{C}$ in $\mathbb{F}_2^7$ are precisely the fibers of the syndrome map

$$\mathbb{F}_2^7 \xrightarrow{H} \mathbb{F}_2^3.$$

Each syndrome labels one coset.

11.3 Single-bit errors

Let $e_i \in \mathbb{F}_2^7$ denote the vector with a 1 in position i and 0 elsewhere. Then

$$He_i^\top$$

is exactly the i-th column of H.

Hence, for this matrix, each single-bit error has a syndrome equal to the corresponding column of H. In Hamming-type constructions, the columns are chosen to be all nonzero vectors of $\mathbb{F}_2^3$, so each nonzero syndrome identifies a unique single-bit error.

12 Conceptual Summary

The story can be summarized as follows.

Finite-field linear algebra

- There are q^{m^2} square $m \times m$ matrices over $\mathbb{F}_q$.
- The invertible ones are counted by

$$|\operatorname{GL}_m(\mathbb{F}_q)| = \prod_{i=0}^{m-1} (q^m - q^i).$$

- Therefore

$$\mathbb{P}(A \in \operatorname{GL}_m(\mathbb{F}_q)) = \prod_{j=1}^{m} (1 - q^{-j}).$$

Rectangular full-rank matrices

- The number of full-row-rank $r \times n$ matrices is

$$\prod_{i=0}^{r-1} (q^n - q^i).$$

- Therefore

$$\mathbb{P}(H \in \mathbb{F}_q^{r\times n} \text{ has rank } r) = \prod_{i=0}^{r-1} (1 - q^{i-n}).$$

Coding theory

- A parity-check matrix $H \in \mathbb{F}_q^{(n-k)\times n}$ defines a code

$$\mathcal{C} = \operatorname{Ker}(H) \subseteq \mathbb{F}_q^n.$$

- If H has full row rank $n-k$, then the syndrome map

$$H : \mathbb{F}_q^n \to \mathbb{F}_q^{n-k}$$

  is surjective.

- Rank-nullity then gives

$$\dim(\mathcal{C}) = k.$$

- The syndrome of a received word y is $Hy^\top$.
- If $y = c + e$ with $c \in \mathcal{C}$, then
$$Hy^\top = He^\top.$$
- Two words have the same syndrome if and only if they lie in the same coset of $\mathcal{C}$.

Exact-sequence form

All of this is neatly encoded by

$$0 \longrightarrow \mathcal{C} \longrightarrow \mathbb{F}_q^n \xrightarrow{H} \mathbb{F}_q^{n-k} \longrightarrow 0.$$

13 Further Remarks

Remark 13.1. The finite-field counting formulas are not just combinatorial curiosities. They explain why a random rectangular matrix is very likely to have full rank when the field size or ambient dimension is reasonably large. This is why random parity-check matrices often define codes of the intended dimension.

Remark 13.2. In coding theory, one often moves freely between three viewpoints on a linear code:

(1) $\mathcal{C}$ as a subspace of $\mathbb{F}_q^n$,

(2) $\mathcal{C}$ as the image of an encoder $G : \mathbb{F}_q^k \to \mathbb{F}_q^n$,

(3) $\mathcal{C}$ as the kernel of a parity-check map $H : \mathbb{F}_q^n \to \mathbb{F}_q^{n-k}$.

The discussion in these notes has emphasized the third viewpoint.

Remark 13.3. The quotient interpretation

$$\mathbb{F}_q^n/\mathcal{C} \cong \mathbb{F}_q^{n-k}$$

is one of the cleanest conceptual bridges from abstract linear algebra to practical syndrome decoding.

Exercises

1. Verify directly that the number of 2×2 invertible matrices over $\mathbb{F}_2$ is 6.
2. Show that the probability a random 4×4 matrix over $\mathbb{F}_3$ is invertible is
$$(1 - 3^{-1})(1 - 3^{-2})(1 - 3^{-3})(1 - 3^{-4}).$$
3. Let $H \in \mathbb{F}_q^{r \times n}$ have full row rank. Prove directly that the fibers of the map
$$H : \mathbb{F}_q^n \to \mathbb{F}_q^r$$
are exactly the cosets of $\mathrm{Ker}(H)$.
4. For the binary matrix
$$H = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix},$$
compute the syndrome of each standard basis vector $e_1, \ldots, e_7$.
5. Prove that if $y = c + e$ with $c \in \mathcal{C} = \mathrm{Ker}(H)$, then the syndrome $Hy^\top$ depends only on the coset $y + \mathcal{C}$.